Hierarchical Pruning of Deep Ensembles with Focal Diversity

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Deep neural network ensembles combine the wisdom of multiple deep neural networks to improve the generalizability and robustness over individual networks. It has gained increasing popularity to study and apply deep ensemble techniques in the deep learning community. Some mission-critical applications utilize a large number of deep neural networks to form deep ensembles to achieve desired accuracy and resilience, which introduces high time and space costs for ensemble execution. However, it still remains a critical challenge whether a small subset of the entire deep ensemble can achieve the same or better generalizability and how to effectively identify these small deep ensembles for improving the space and time efficiency of ensemble execution. This paper presents a novel deep ensemble pruning approach, which can efficiently identify smaller deep ensembles and provide higher ensemble accuracy than the entire deep ensemble of a large number of member networks. Our hierarchical ensemble pruning approach (HQ) leverages three novel ensemble pruning techniques. First, we show that the focal ensemble diversity metrics can accurately capture the complementary capacity of the member networks of an ensemble team, which can guide ensemble pruning. Second, we design a focal ensemble diversity based hierarchical pruning approach, which will iteratively find high quality deep ensembles with low cost and high accuracy. Third, we develop a focal diversity consensus method to integrate multiple focal diversity metrics to refine ensemble pruning results, where smaller deep ensembles can be effectively identified to offer high accuracy, high robustness and high ensemble execution efficiency. Evaluated using popular benchmark datasets, we demonstrate that the proposed hierarchical ensemble pruning approach can effectively identify high quality deep ensembles with better classification generalizability while being more time and space efficient in ensemble decision making. We have released the source codes on GitHub at https://github.com/git-disl/HQ-Ensemble.

CCS Concepts: • **Computing methodologies → Ensemble methods**.

Additional Key Words and Phrases: Ensemble Pruning, Ensemble Learning, Ensemble Diversity, Deep Learning

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1 INTRODUCTION

It has become an attractive learning technique to leverage deep neural network (DNN) ensembles to improve the overall generalizability and robustness of many deep learning systems. Some mission-critical applications often require a large number of deep neural networks to achieve the target accuracy and robustness, which entails high space and time costs for ensemble execution. Recent

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studies have revealed that deep neural network ensembles with highly diverse member networks tend to have high failure independence, which is critical for enhancing overall ensemble predictive performance, including ensemble accuracy and robustness under adverse situations [20, 34, 36, 38, 39]. However, the member networks in a large deep ensemble team may not have high ensemble diversity and failure independence to complement each other, which will result in sub-optimal ensemble prediction performance and high ensemble execution cost in practice [10, 20, 35, 37–39]. For a deep ensemble team of a large size, such as 10, it is often not only possible to identify substantially smaller deep ensembles (e.g., 3∼5 member networks) with the same or improved ensemble accuracy but also beneficial to reduce the ensemble execution cost [18, 24, 37–39]. This motivates us to propose an efficient hierarchical ensemble pruning approach, coined as HQ. By leveraging our focal diversity metrics, the HQ pruning method can effectively identify high quality deep ensembles with small sizes and high ensemble diversity, which not only improves ensemble accuracy over the large entire ensembles but also significantly reduces the space and time cost of ensemble execution.

1.1 Related Work and Problem Statement

Most of the existing ensemble learning studies can be summarized into three broad categories. The first category builds ensemble teams by training a set of member models, represented by bagging [4], boosting [6] and random forests [5]. In practice, it has led to large ensemble teams of tens to hundreds of member models, such as the commonly-used random forests. The second category is to leverage diversity measurements to compare and select high quality ensemble teams whose member models can complement each other to improve the ensemble predictive performance without requiring model training. Ensemble diversity can be evaluated using pairwise or non-pairwise diversity measures, represented by Cohen's Kappa (CK) [25] and Binary Disagreement (BD) [30] for pairwise metrics and Kohavi-Wolpert Variance (KW) [15, 17] and Generalized Diversity (GD) [27] for non-pairwise metrics. Early studies [17, 31] have reported that it is challenging to use these diversity metrics for evaluating the performance quality of ensemble teams. Some recent studies [10, 20, 38, 39] discussed inherent problems of using these diversity metrics to measure ensemble diversity in terms of failure independence and provided guidelines for improving the ensemble diversity measurement. This paper contributes to this second category by leveraging focal ensemble diversity metrics for effective pruning of deep ensembles. The third category covers the ensemble consensus methods for aggregating member model predictions and producing the ensemble prediction, such as soft voting (model averaging), majority voting and plurality voting [13] or learn to combine algorithms [21, 38].

Table 1. Summary of Three Categories of Related Studies

Category	Description	Representative methods
1. Ensemble training	Training a set of member models to build ensemble teams	Bagging [4], boosting [6], random forests [5], etc.
2. Ensemble diversity powered ensemble pruning	Leveraging ensemble diversity measurements to compare and select high-quality ensembles	CK pruning [25], BD pruning [30], GD pruning [27], etc.
3. Ensemble consensus methods	Aggregating member model predictions to produce the ensemble prediction	Soft voting, majority voting, plurality voting, etc.

We summarize the three categories of related studies in Table 1. These three categories are complementary. To develop an ensemble team for improving prediction performance, we can employ one of the two ways to obtain an ensemble of member models: (1) training multiple models together using an ensemble learning algorithm, such as bagging, boosting, or random forest; and (2) training multiple models independently in parallel and employ a voting method to produce

Table 2. Comparison of Hierarchical Pruning and Existing Studies

Method	Model Types	Diversity Measurement Principles		
		Diversity Comparison	Samples for Measurements	Diversity Calculation
Early studies on diversity based ensemble pruning before 2015 [2, 7, 18, 24, 32]	Ensembles of traditional ML models	Compare all ensembles of different sizes	Random samples from the validation set	Directly calculated on random samples
Focal diversity powered hierarchical ensemble pruning	Ensembles of (1) DNNs and (2) Traditional ML models	Compare the ensembles of the same size S	Random negative samples from a focal model F_f	Obtain focal negative correlations for each focal (member) model F_f in an ensemble team of size S and then perform an average of S focal negative correlation scores to calculate the focal diversity score

ensemble consensus based predictions. Most of the early studies before 2015 in the first two categories [2, 7, 18, 24, 32] focused on ensemble pruning for traditional machine learning models. Until recently, we have observed several research endeavors on deep neural network ensembles, most of which centered on training multiple networks jointly, such as diversity based weighted kernels [3, 8, 40] and leveraging deep ensembles to strengthen the robustness and resilience of a single deep neural network under adverse situations [8, 9, 20, 34–36, 39]. However, to the best of our knowledge, very few studies have brought forward solutions to efficient deep ensemble pruning to improve prediction performance and reduce ensemble execution cost.

We compare our focal diversity powered hierarchical ensemble pruning approach and early studies using ensemble diversity for ensemble pruning in Table 2. Our focal diversity promotes fair comparison of ensemble diversity among the ensembles of the same size S and leverages the focal model concept, focal negative correlation, and averaging of multiple focal negative correlation scores to obtain accurate ensemble diversity measurements. We will provide a detailed description of focal diversity powered hierarchical pruning in Section 3 and demonstrate that our hierarchical pruning approach can be effectively applied to both deep neural network ensembles and ensembles of traditional machine learning models in Section 5.

1.2 Scope and Contribution

This paper presents a holistic approach to efficient deep ensemble pruning. Given a large deep ensemble of M member networks, we propose a hierarchical ensemble pruning framework, denoted as HQ, to efficiently identify high quality ensembles with lower cost and higher accuracy than the entire ensemble of M networks. Our HQ framework combines three novel ensemble pruning techniques. *First,* we leverage focal ensemble diversity metrics [39] to measure the failure independence among member networks of a deep ensemble. The higher focal diversity score indicates the higher level of failure independence among member networks of a deep ensemble, that is higher complementary capacity for improving ensemble predictions. Our focal diversity metrics can precisely capture such correlations among member networks of an ensemble team, which can be used to effectively guide ensemble pruning. *Second,* we present a novel hierarchical ensemble pruning method powered by focal diversity metrics. Our hierarchical pruning approach iteratively identifies subsets of member networks with low diversity, which tend to make similar prediction errors, and then prunes them out from the entire ensemble team. *Third,* we perform focal diversity consensus voting to combine multiple focal diversity metrics for ensemble pruning, which further refines the hierarchical ensemble pruning results by a single focal diversity metric. Comprehensive experiments are performed on four popular benchmark datasets, CIFAR-10 [16], ImageNet [29],